

A Parametric Approach to Quantifying Risk & Uncertainty in Indian Market-Linked Debentures

Group 8

Project Mentor - Prof. Vaibhav Vasundekar



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Introduction

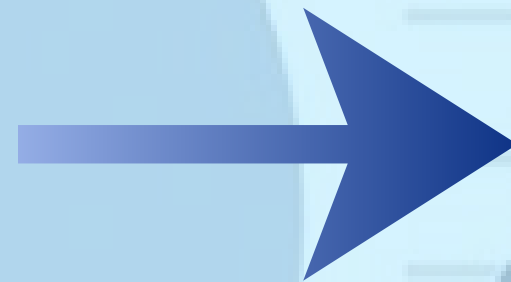
Objectives & Workflow

Methodology

Conclusions

Limitations

Future Scopes



1. Computing Log Returns
2. Exploratory Data Analysis (EDA)
3. Volatility Modelling (eGARCH)
4. Standardized Residual Analysis
5. Distribution Fitting (MLE)
6. Tail Modelling & Goodness of Fit
7. Risk Measures (VaR & CTE)
8. Application to MLD Payoff

Introduction

- Investor Choices range from direct investment to equity-linked investments.
- Direct Investments- High returns, High volatility and unsystematic risks
- Indirect Investments- Capped returns and complex risk structures
- Compare and quantify risks linked to both cases

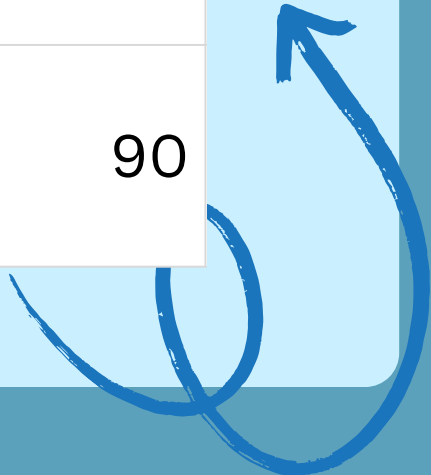


1. PRINCIPAL-PROTECTED MLD:

Nifty 50 Performance Over 5 Years	Investor's Return	Amount Received at Maturity (₹)
Rises by 20%	15% return	115
Rises by 10%	10% return	110
No Change	0% return	100
Falls by 15%	0% return	100

2. NON-PRINCIPAL-PROTECTED MLD:

Nifty 50 Performance Over 5 Years	Investor's Return	Amount Received at Maturity (₹)
Rises by 20%	15% return	115
Rises by 10%	10% return	110
No Change	0% return	100
Falls by 15%	-10% loss	90



Data collection

Stock indices collected :

- Nifty 50
- Banknifty
- Sensex

Return calculation: We need to model the distribution of the returns of the stocks. Hence, we calculate the log-returns

$$R_t = \log \left(\frac{P_t}{P_{t-1}} \right)$$

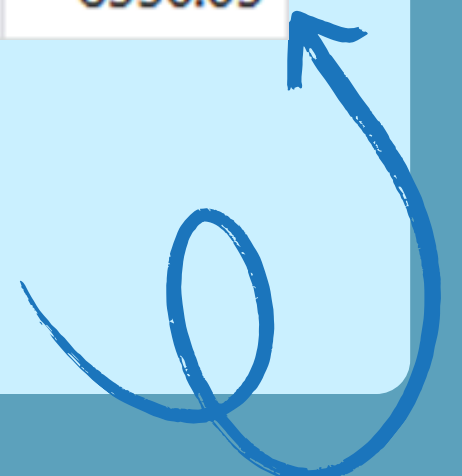
Where,

R_t : return value at time t

P_t : stock closing price at time t

P_{t-1} : stock closing price at time $(t-1)$

	▲	◀	◀	◀	◀	◀
	Date	Adj Close	Close	High	Low	Open
1	2015-01-05	8378.40	8378.40	8445.60	8363.90	8407.95
2	2015-01-06	8127.35	8127.35	8327.85	8111.35	8325.30
3	2015-01-07	8102.10	8102.10	8151.20	8065.45	8118.65
4	2015-01-08	8234.60	8234.60	8243.50	8167.30	8191.40
5	2015-01-09	8284.50	8284.50	8303.30	8190.80	8285.45
6	2015-01-12	8323.00	8323.00	8332.60	8245.60	8291.35
7	2015-01-13	8299.40	8299.40	8356.65	8267.90	8346.15
8	2015-01-14	8277.55	8277.55	8326.45	8236.65	8307.25
9	2015-01-15	8494.15	8494.15	8527.10	8380.55	8424.50
10	2015-01-16	8513.80	8513.80	8530.75	8452.25	8504.05
11	2015-01-19	8550.70	8550.70	8570.95	8531.50	8550.05



Objectives & Workflow

Model Stock returns

- Computing Log Returns
- Exploratory Data Analysis (EDA)
- Volatility Modelling (eGARCH)
- Standardized Residual Analysis

Optimal Model Selection

- Distribution Fitting (MLE)
- Tail Modelling & Goodness of Fit AIC, BIC , K-S test

Risk Analysis

- Value at risk (VaR)
- Conditional Tail Expectation (CTE)
- Model Accuracy (Backtesting)

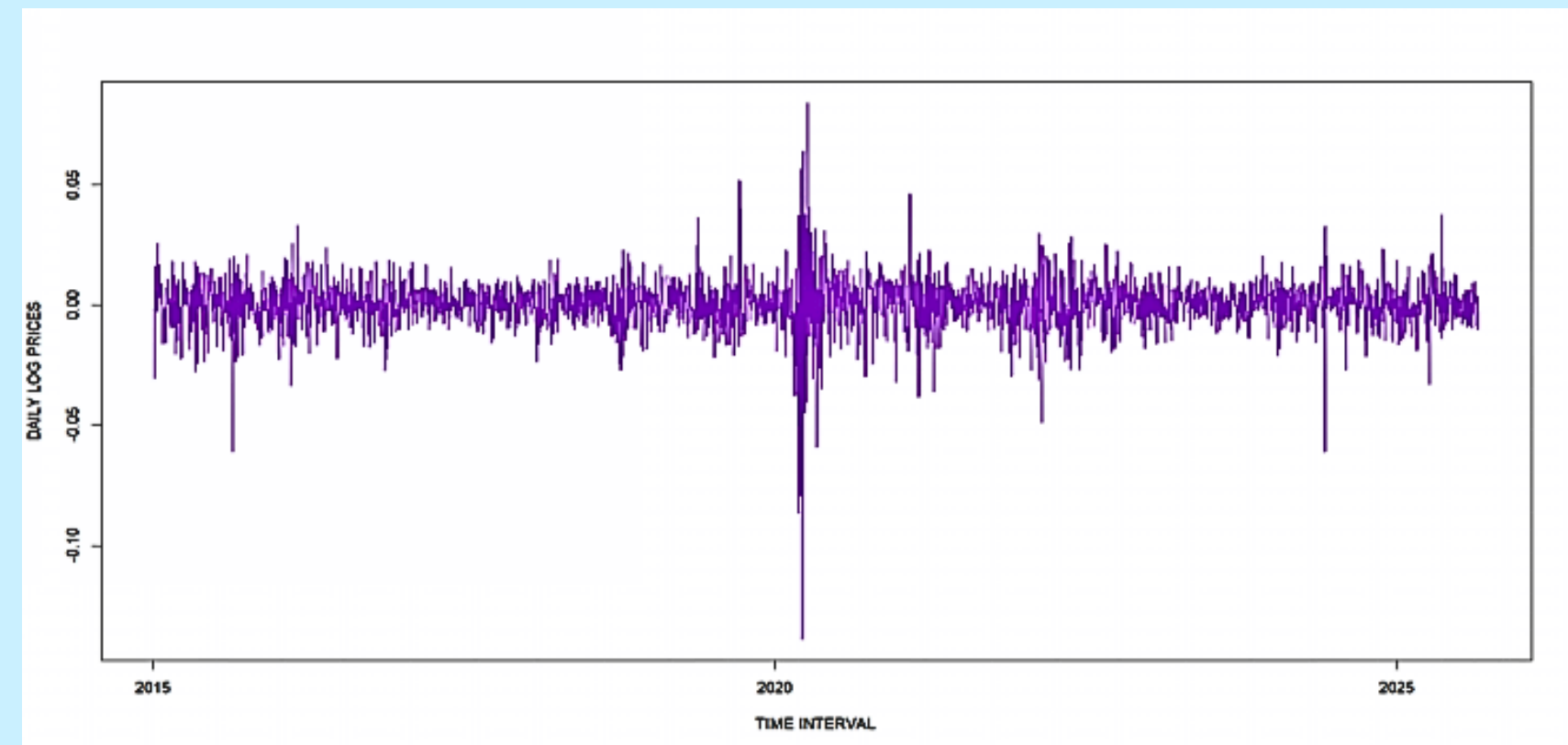
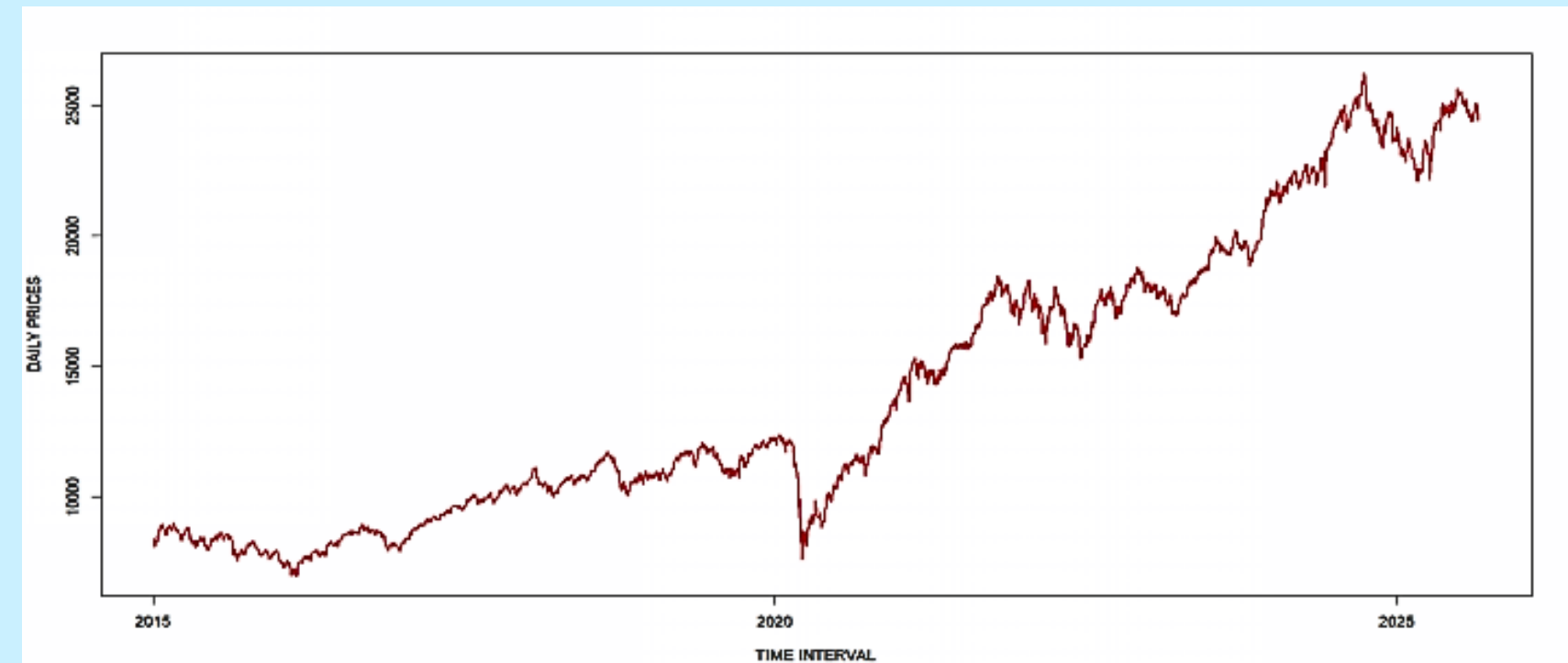
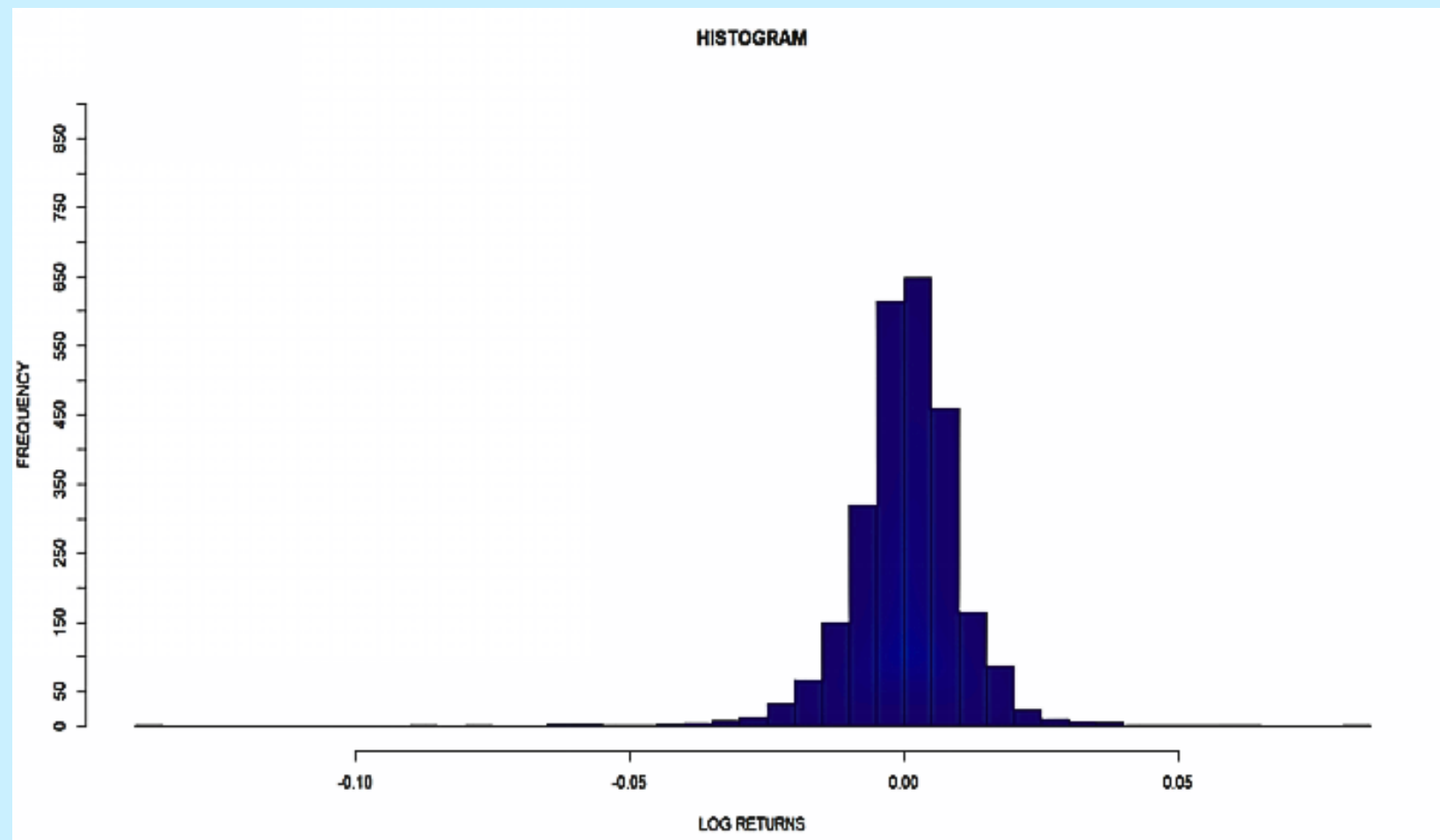
Comparative Analysis

- Direct Investments
- MLDs Principal Protected
- MLDs Non-Principal Protected MLDs

Methodology

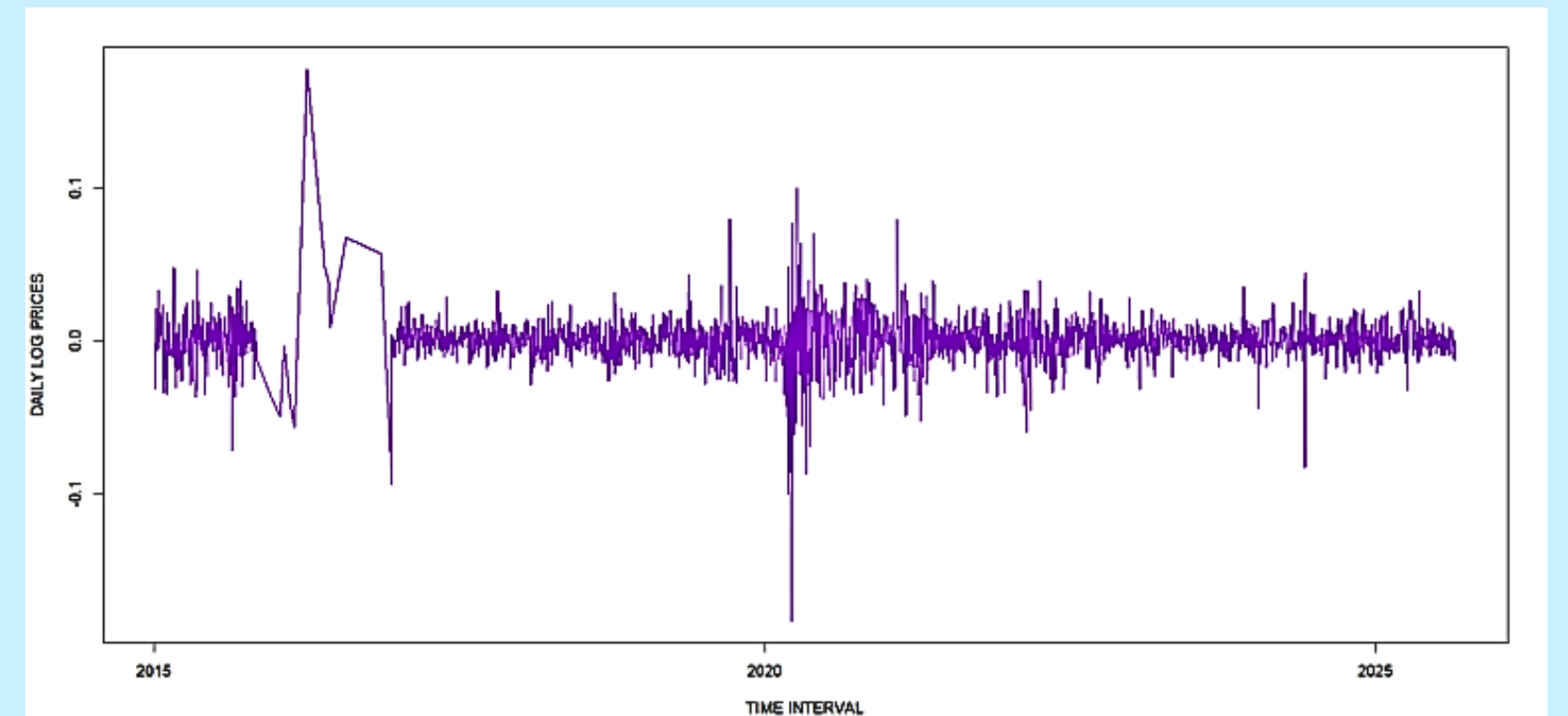
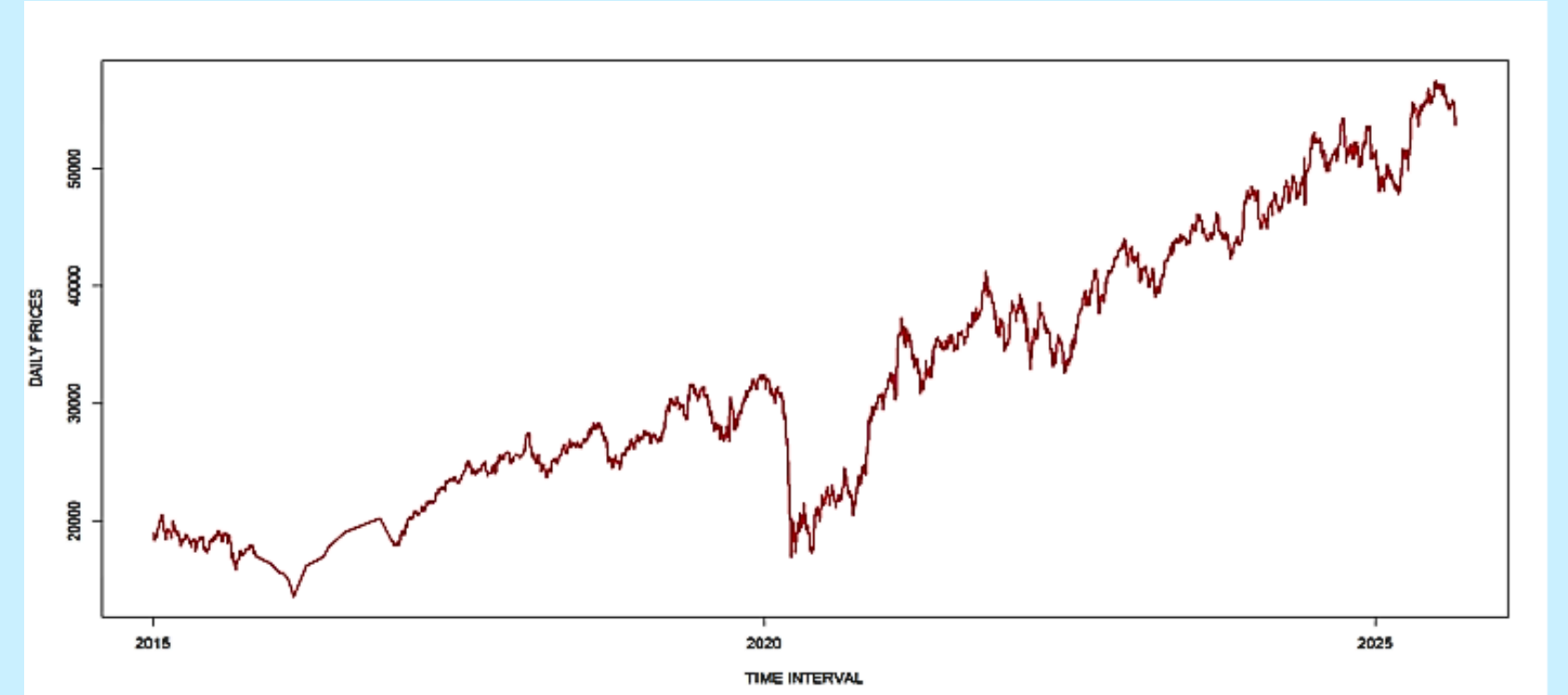
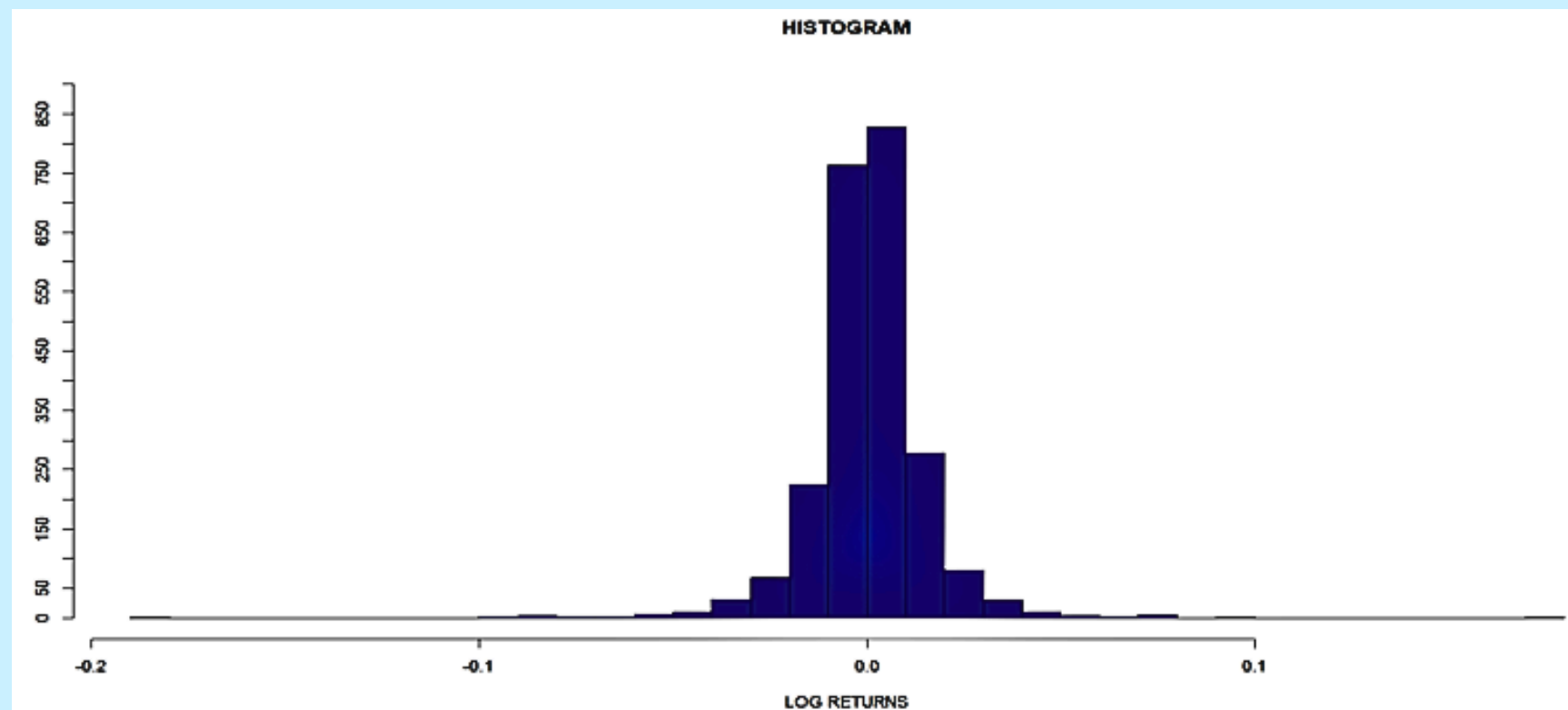
2. Exploratory data analysis

A) For Nifty 50



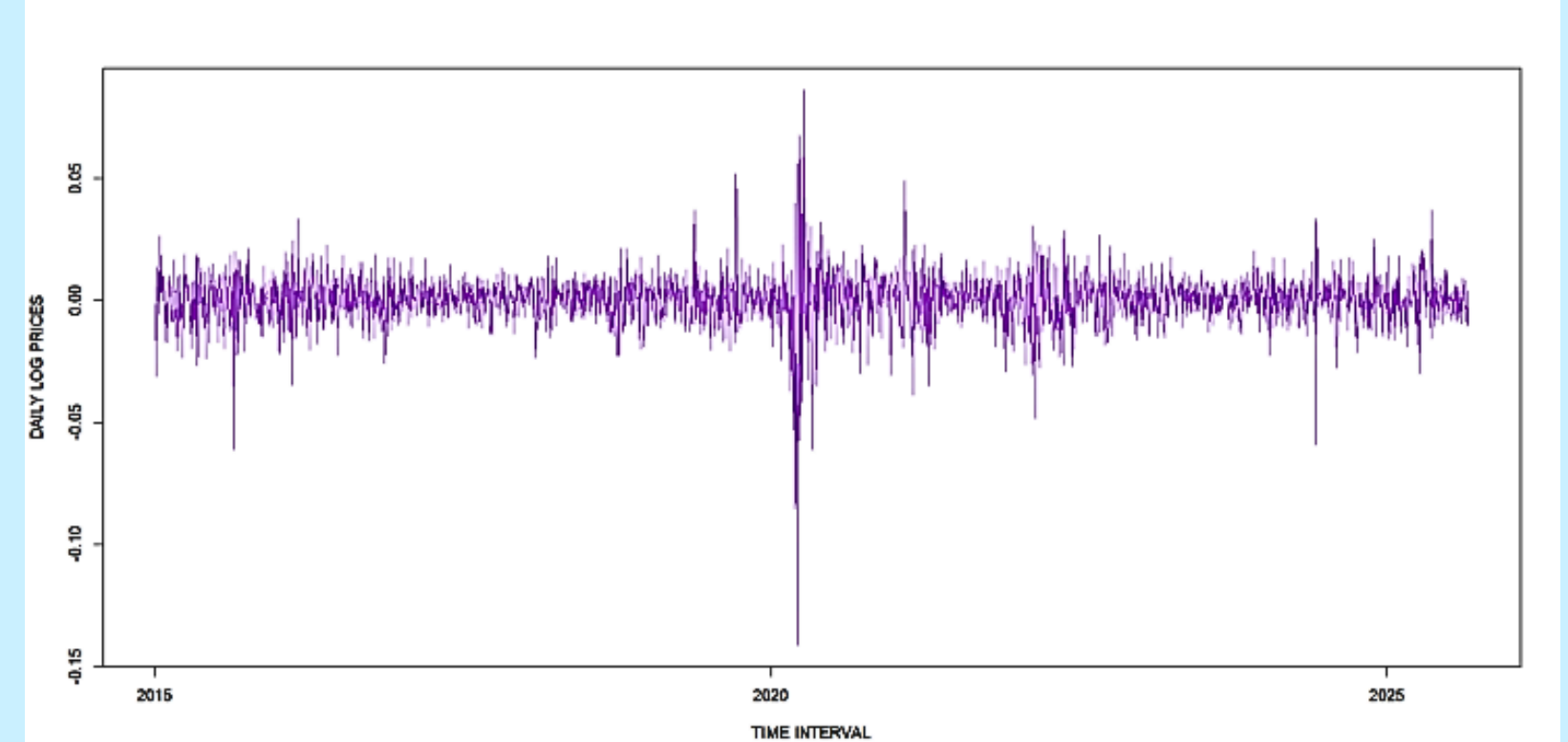
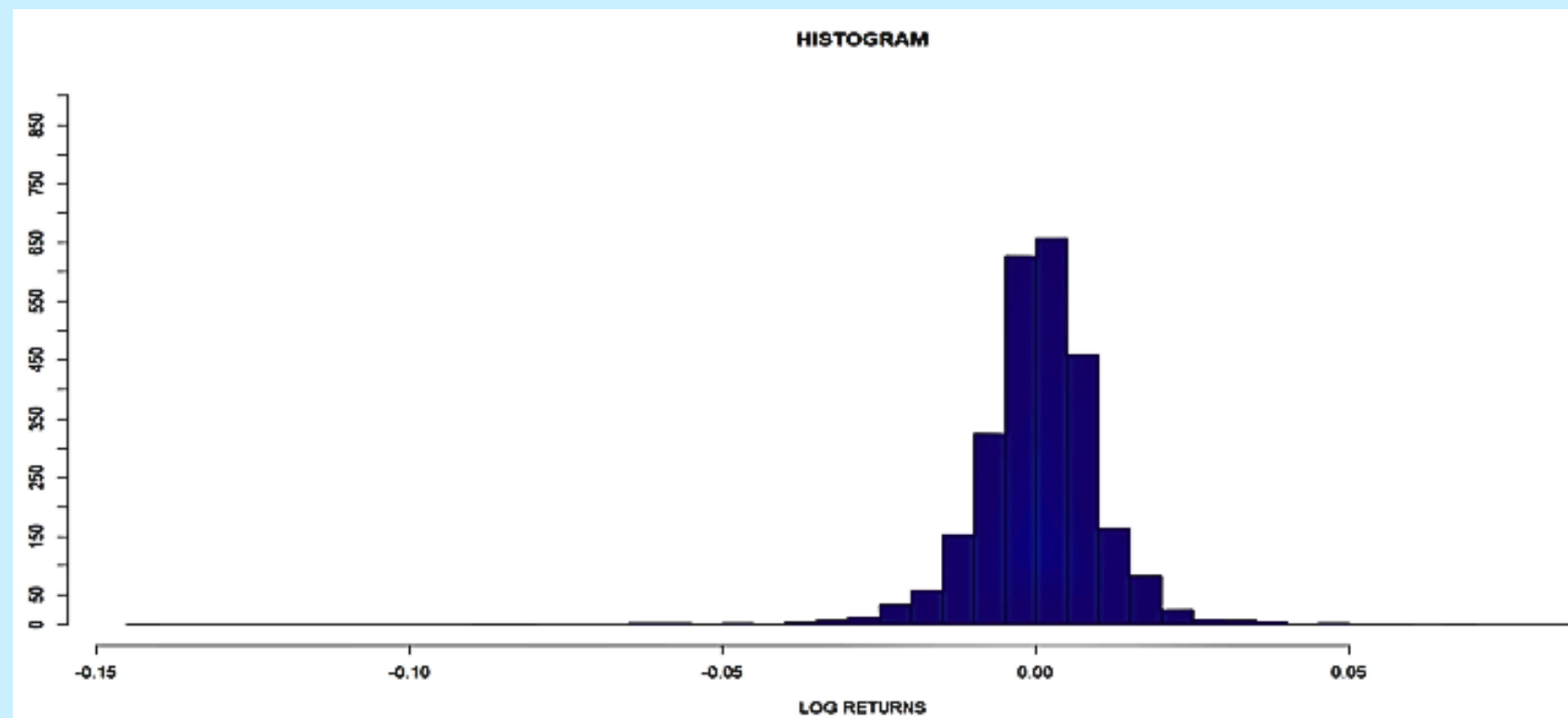
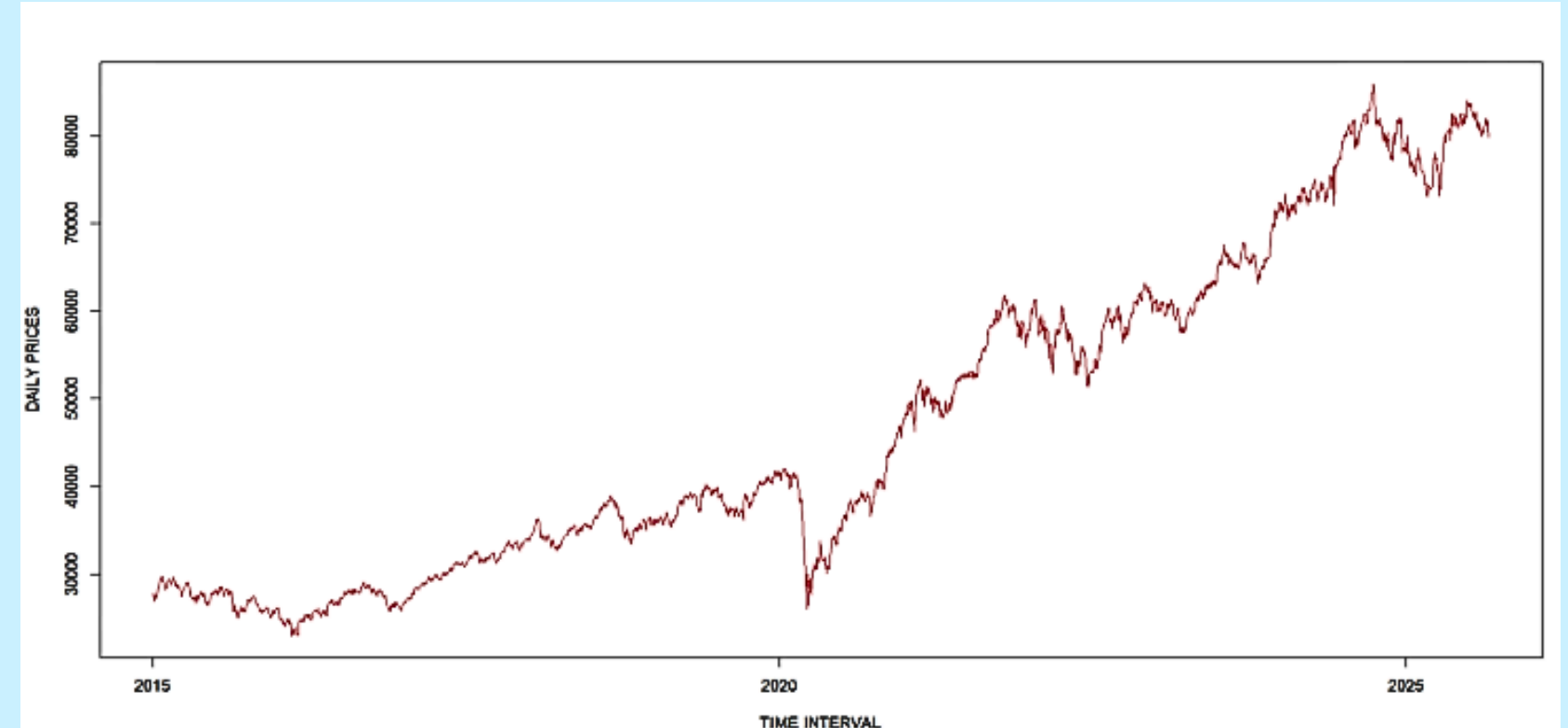
Methodology

2. Exploratory data analysis B) For Banknifty



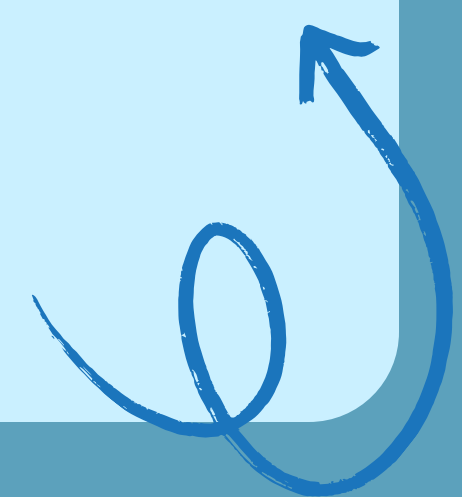
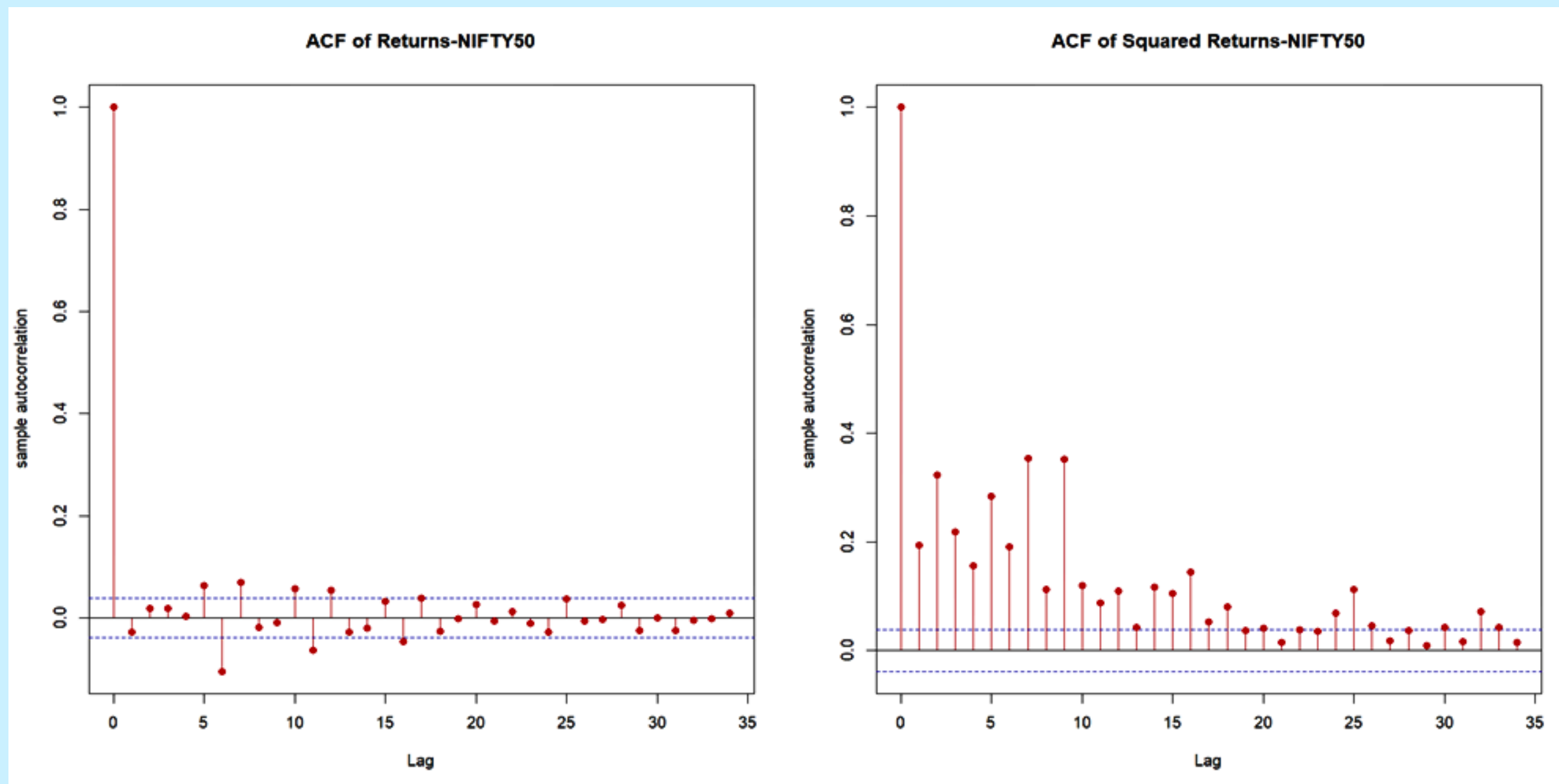
Methodology

2. Exploratory data analysis C) For Sensex



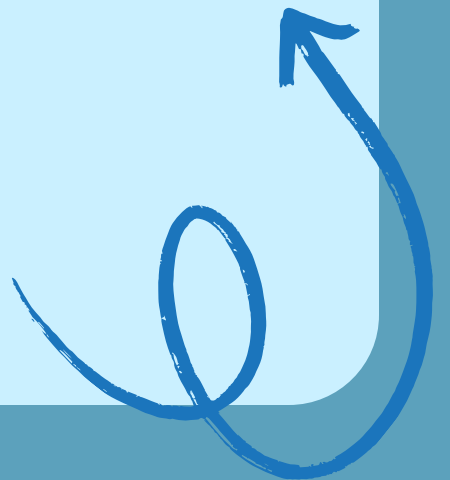
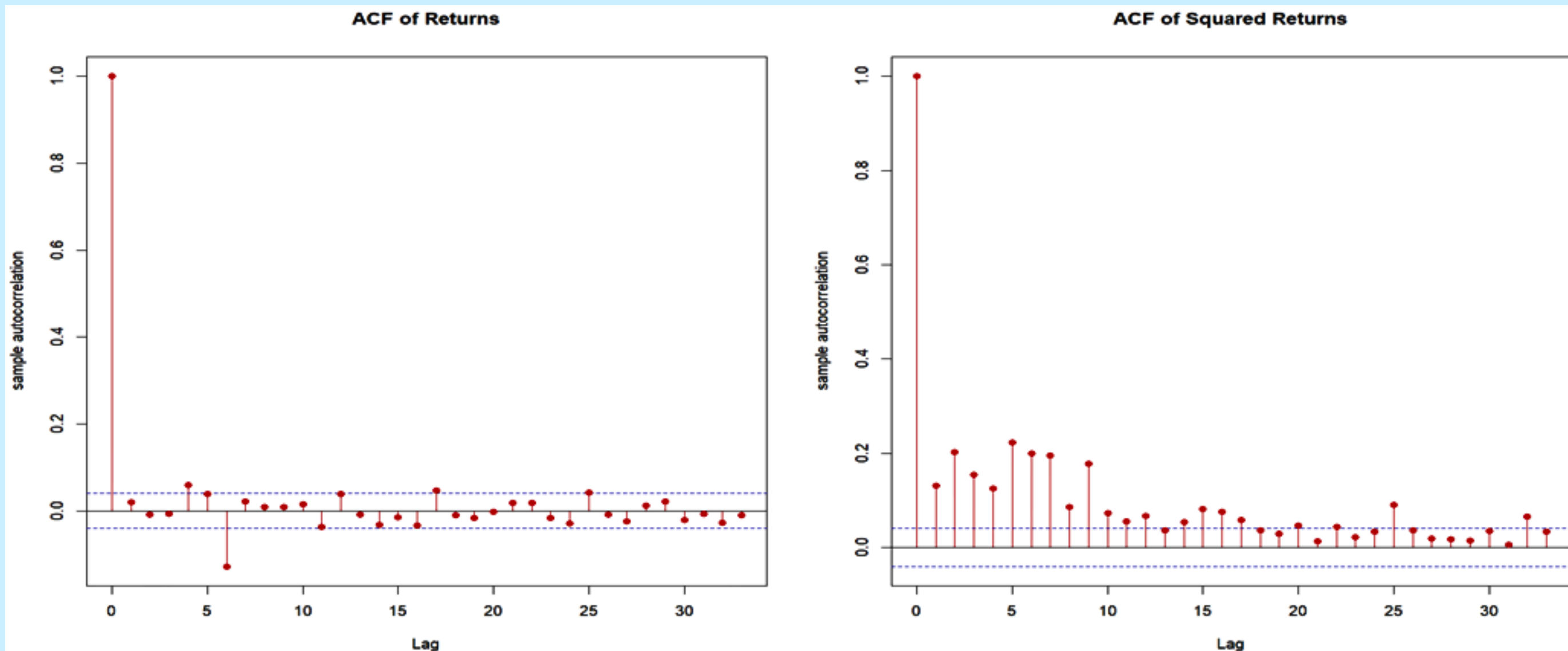
Methodology

2. Exploratory Data Analysis ACF plots for NIFTY-50



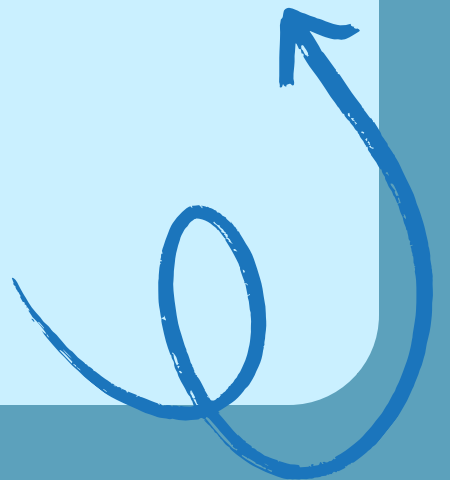
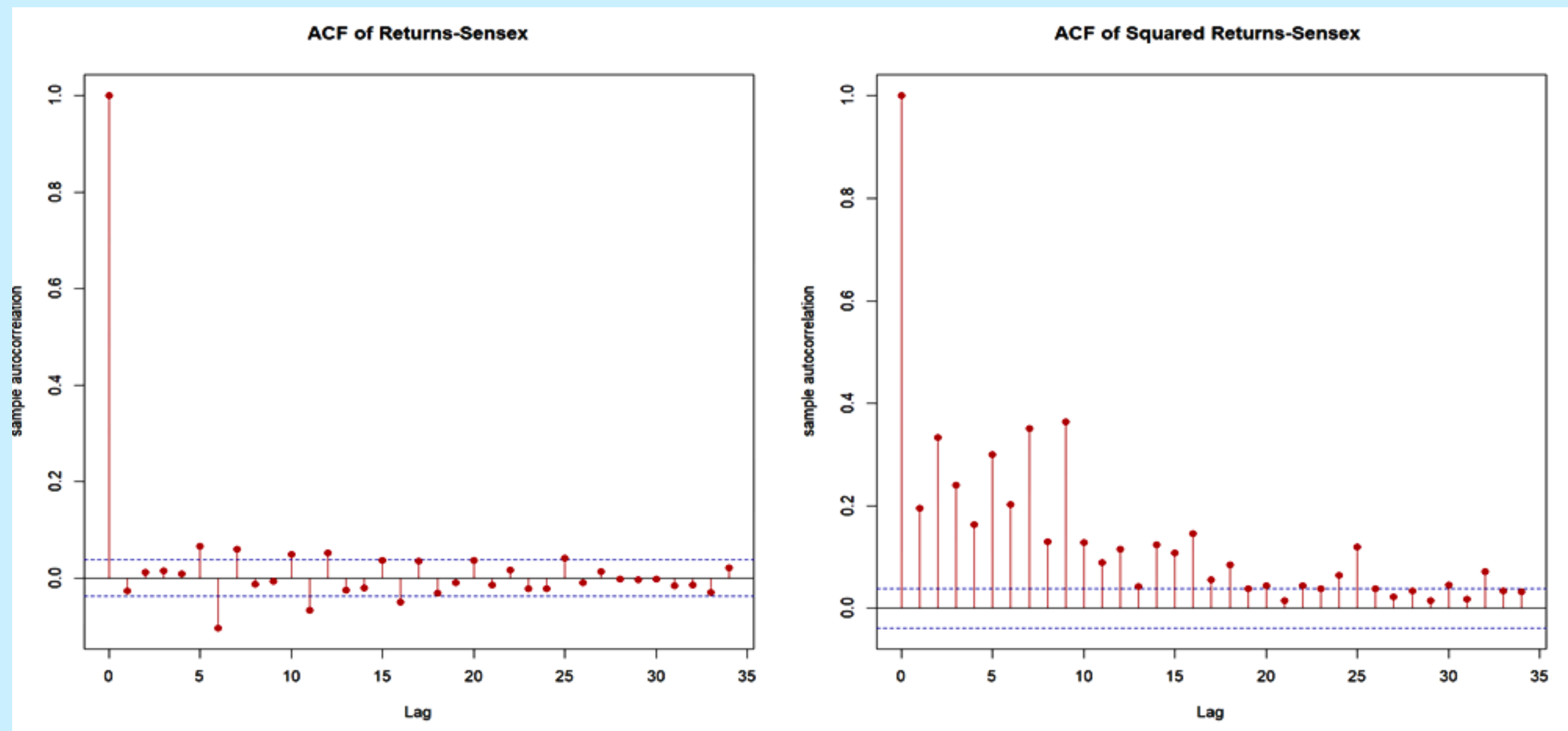
Methodology

2. Exploratory Data Analysis ACF plots for Banknifty



Methodology

2. Exploratory Data Analysis ACF plots for Sensex



Methodology

3. Volatility Modelling (eGARCH)

In our study, we found that

1. The log returns and squared log returns are highly correlated
2. Hence, we used ARMA(1,1)-egarch(1,1) model under skew-t innovations.
3. This makes the model approximately independent and identically distributed.

- **Conditional Mean Equation:**

$$r_t = \mu_t + \epsilon_t$$

- **Conditional Variance Equation (egarch(1,1)):**

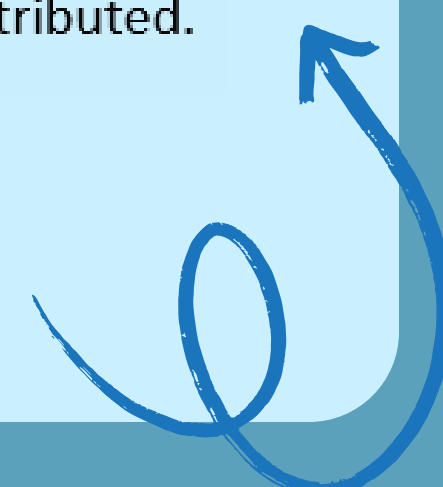
$$\log(\sigma_t^2) = \omega + \alpha \left(\frac{|\epsilon_{t-1}|}{\sigma_{t-1}} - E \left[\frac{|\epsilon_{t-1}|}{\sigma_{t-1}} \right] \right) + \gamma \frac{\epsilon_{t-1}}{\sigma_{t-1}} + \beta \log(\sigma_{t-1}^2)$$

- **Filtering the Returns:**

We create a filtered return series z_t as follows

$$z_t = \frac{\epsilon_t}{\sigma_t} = \frac{r_t - \mu_t}{\sigma_t}$$

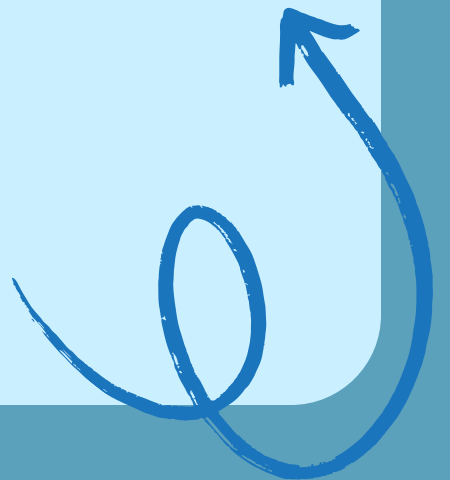
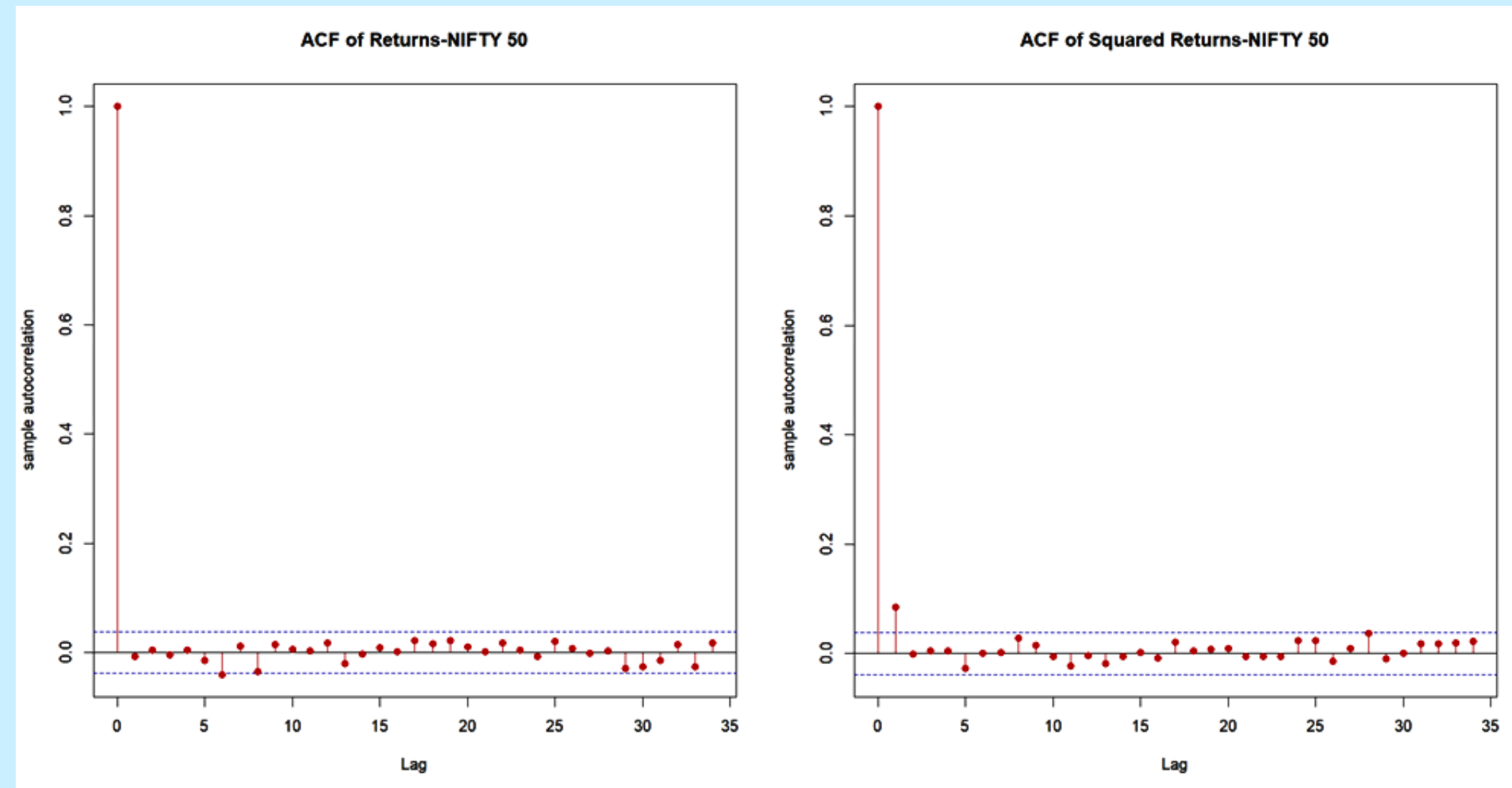
z_t should be approximately independent and identically distributed.



Methodology

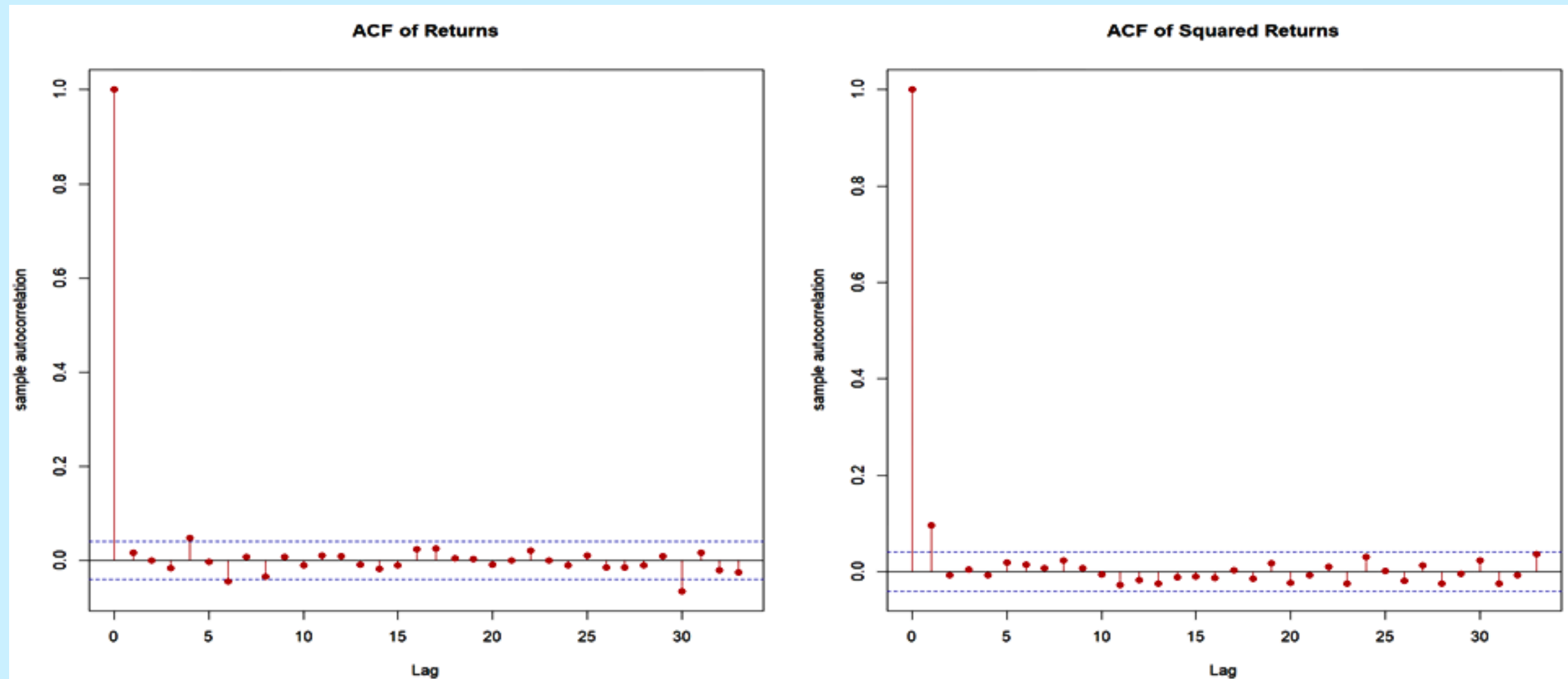
4. Standardized Residual Analysis

A. Nifty 50



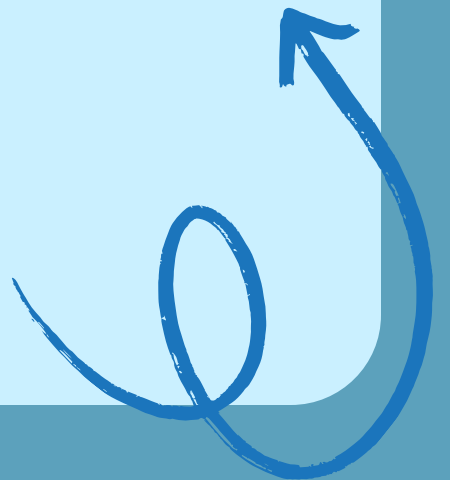
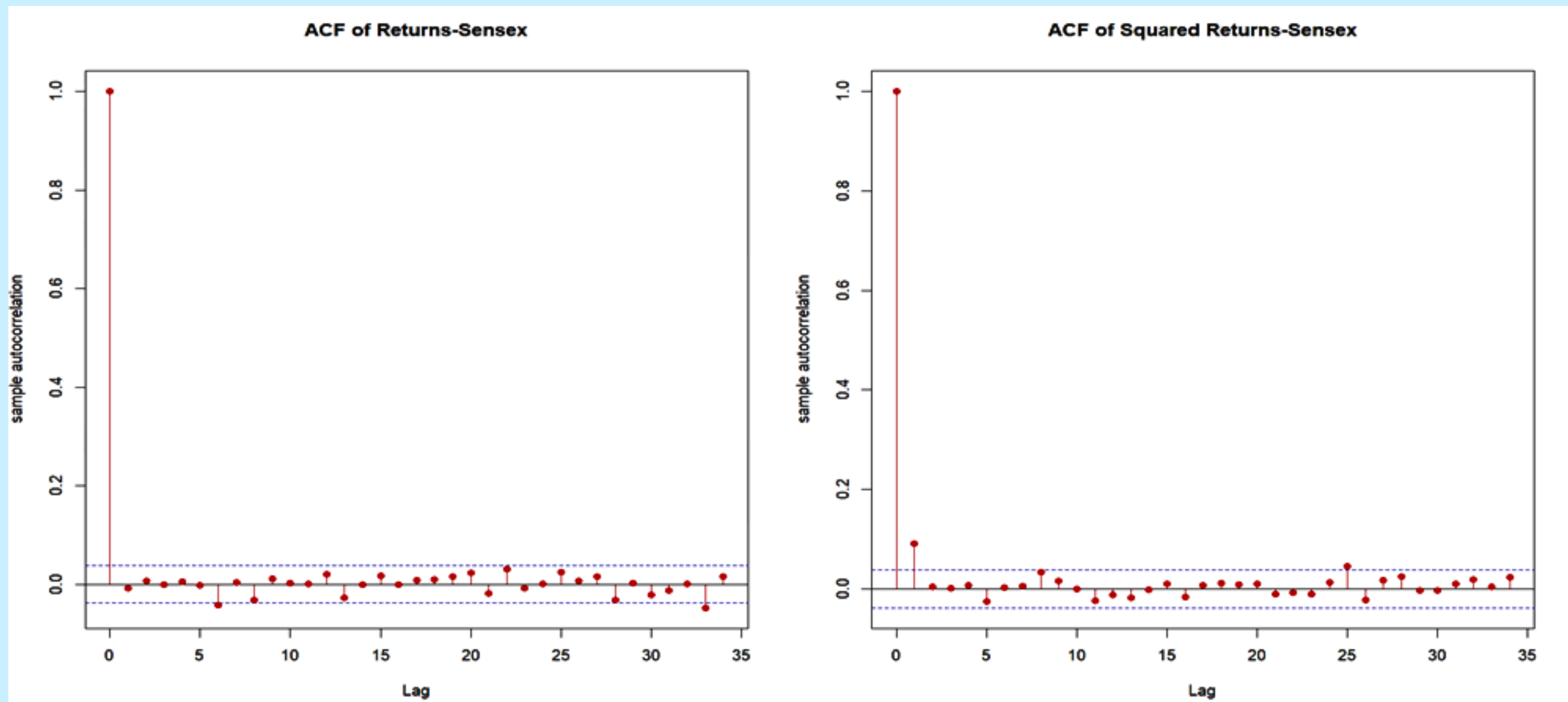
Methodology

4. Standardized Residual Analysis B. Banknifty



Methodology

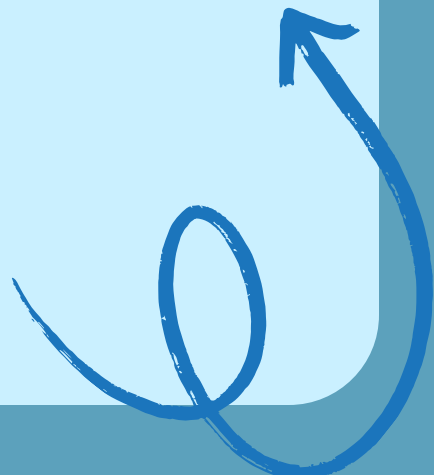
4. Standardized Residual Analysis C. Sensex



Methodology

5. Distribution Fitting (MLE) - NIFTY 50

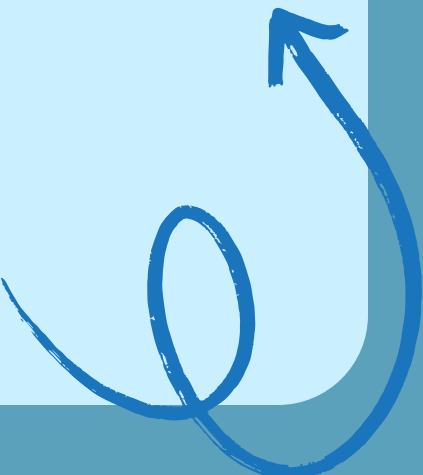
Model	Parameters					Log-Likelihood
Normal	μ -0.001643094	σ 1.008374884				-3743.752
Laplace	μ 0.02589474	σ 0.7599479				-3721.097
Cauchy	μ 0.05333791	σ 0.55351445				-4020.997
Student's t	μ 0.03191978	σ 0.83779143	ν 6.66094094			-3664.564
Skew Normal	μ -0.008552	σ 1.0023	α -0.2964			-3706.151
Skew t	μ 0.5503128	σ 0.9710637	α -0.8188604	ν 7.1830738		-3651.382
Skew Cauchy	μ 0.1504727	σ 0.5591503	α -0.1682965			-4018.304
Skew Laplace	μ 0.1202428	σ 0.3768044	α 0.5396423			-3715.666
Hyperbolic	μ 0.3543193	σ 0.9722877	α 2.1835993	β -0.3559412		-3656.839
NIG	μ 0.338164	σ 0.9744375	α 2.3097136	β -0.3398396		-3654.764
Variance Gamma	μ 0.3590726	σ 0.9727949	α 2.9161088	β -0.3609983		-3660.434
G. Hyperbolic	μ 0.3282408	σ 0.9706962	α 0.3553765	β -0.3317509	λ -3.7582256	-3651.33



Methodology

5. Distribution Fitting (MLE) - BANKNIFTY

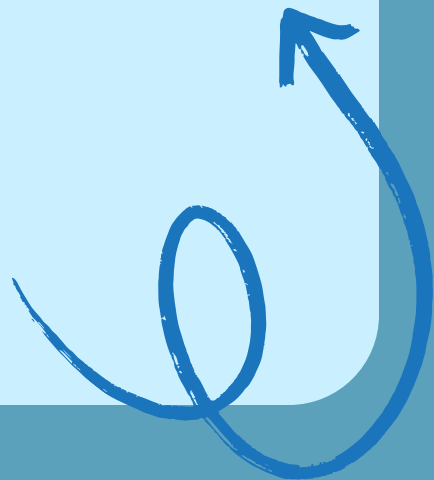
Model	Parameters					Log-Likelihood
Normal	μ 0.002138571	σ 1.001612553				-3334.03
Laplace	μ 0.02518736	σ -0.30152593				-3266.135
Cauchy	μ 0.02990741	σ 0.52694983				-3514.133
Student's <i>t</i>	μ 0.02029246	σ 0.78347466	ν 5.1585773			-3231.475
Skew Normal	μ -0.002056149	σ 0.998600513	α -0.161027361			-3321.059
Skew <i>t</i>	μ 0.002837675	σ 1.000881561	α -0.293909414	ν 5.115471697		-3229.426
Skew Cauchy	μ 0.08021851	σ 0.52794045	α -0.09261486			-3513.34
Skew Laplace	μ 0.07201351	σ 0.3688099	α 0.5229751			-3264.522
Hyperbolic	μ 0.1104484	σ 0.9840238	α 1.178022	β -0.1084271		-3234.988
NIG	μ 0.1065187	σ 0.989541	α 1.390647	β -0.1037641		-3231.93
Variance Gamma	μ 0.1136303	σ 0.9833722	α 0	β -0.1114982		-3237.906
G. Hyperbolic	μ 0.1007881	σ 0.9963472	α 1.106071	β -0.09833108	λ -1.838599	-3230.176



Methodology

5. Distribution Fitting (MLE) - SENSEX

Model	Parameters					Log-Likelihood
Normal	μ -0.001643094	σ 1.008374884				-3743.752
Laplace	μ 0.02589474	σ 0.7599479				-3721.097
Cauchy	μ 0.05333791	σ 0.55351445				-4020.997
Student's <i>t</i>	μ 0.03191978	σ 0.83779143	ν 6.66094094			-3664.564
Skew Normal	μ -0.008552	σ 1.0023	α -0.2964			-3706.151
Skew <i>t</i>	μ 0.5503128	σ 0.9710637	α -0.8188604	ν 7.1830738		-3651.382
Skew Cauchy	μ 0.1504727	σ 0.5591503	α -0.1682965			-4018.304
Skew Laplace	μ 0.1202428	σ 0.3768044	α 0.5396423			-3715.666
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NIG	μ 0.338164	σ 0.9744375	α 2.3097136	β -0.3398396		-3654.764
Variance Gamma	μ 0.3590726	σ 0.9727949	α 2.9161088	β -0.3609983		-3660.434
G. Hyperbolic	μ 0.3282408	σ 0.9706962	α 0.3553765	β -0.3317509	λ -3.7582256	-3651.33



Methodology

6. Tail Modelling and Goodness of fit (AIC , BIC)

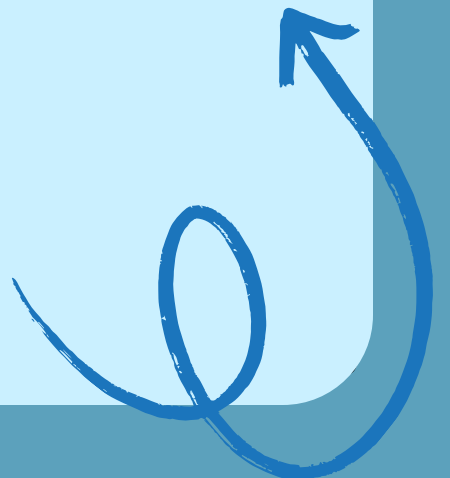
1. Akaike Information Criterion (AIC):

2. Bayesian Information Criterion (BIC):

$$AIC = 2 * K - 2\log L(\hat{\theta})$$

$$BIC = K * \log(n) - 2\log L(\hat{\theta})$$

Model	Nifty50	Banknifty	Sensex
Normal	(7491.503, 7503.247)	(6672.061, 6683.582)	(7537.805, 7549.562)
Laplace	(7446.194, 7457.938)	(6536.269, 6547.791)	(7495.567, 7507.325)
Cauchy	(8045.994, 8057.738)	(7032.266, 7043.788)	(8096.265, 8108.022)
Student's t	(7335.129, 7352.745)	(6468.950, 6486.233)	(7380.496, 7398.131)
Skew normal	(7418.301, 7435.918)	(6648.119, 6665.402)	(7472.401, 7490.037)
Skew t	(7310.764, 7334.252)	(6466.851, 6489.895)	(7361.634, 7385.148)
Skew Cauchy	(8042.607, 8060.224)	(7032.680, 7049.962)	(8094.069, 8088.069)
Skew Laplace	(7437.333, 7454.949)	(6535.045, 6552.327)	(7404.038, 7427.552)
Hyperbolic	(7321.678, 7345.167)	(6477.976, 6501.020)	(7372.157, 7403.550)
NIG	(7317.529, 7341.017)	(6471.860, 6494.903)	(7367.794, 7399.187)
Variance gamma	(7328.868, 7352.357)	(6483.812, 6506.856)	(7379.286, 7410.679)
Generalized	(7312.661, 7342.021)	(6470.352, 6499.157)	(7366.141, 7395.534)



Methodology

6. Tail Modelling and Goodness of fit (KS Test)

3.Kolmogorov Smirnov Test:

$$D = sup| F_n(x) - F(x)|$$

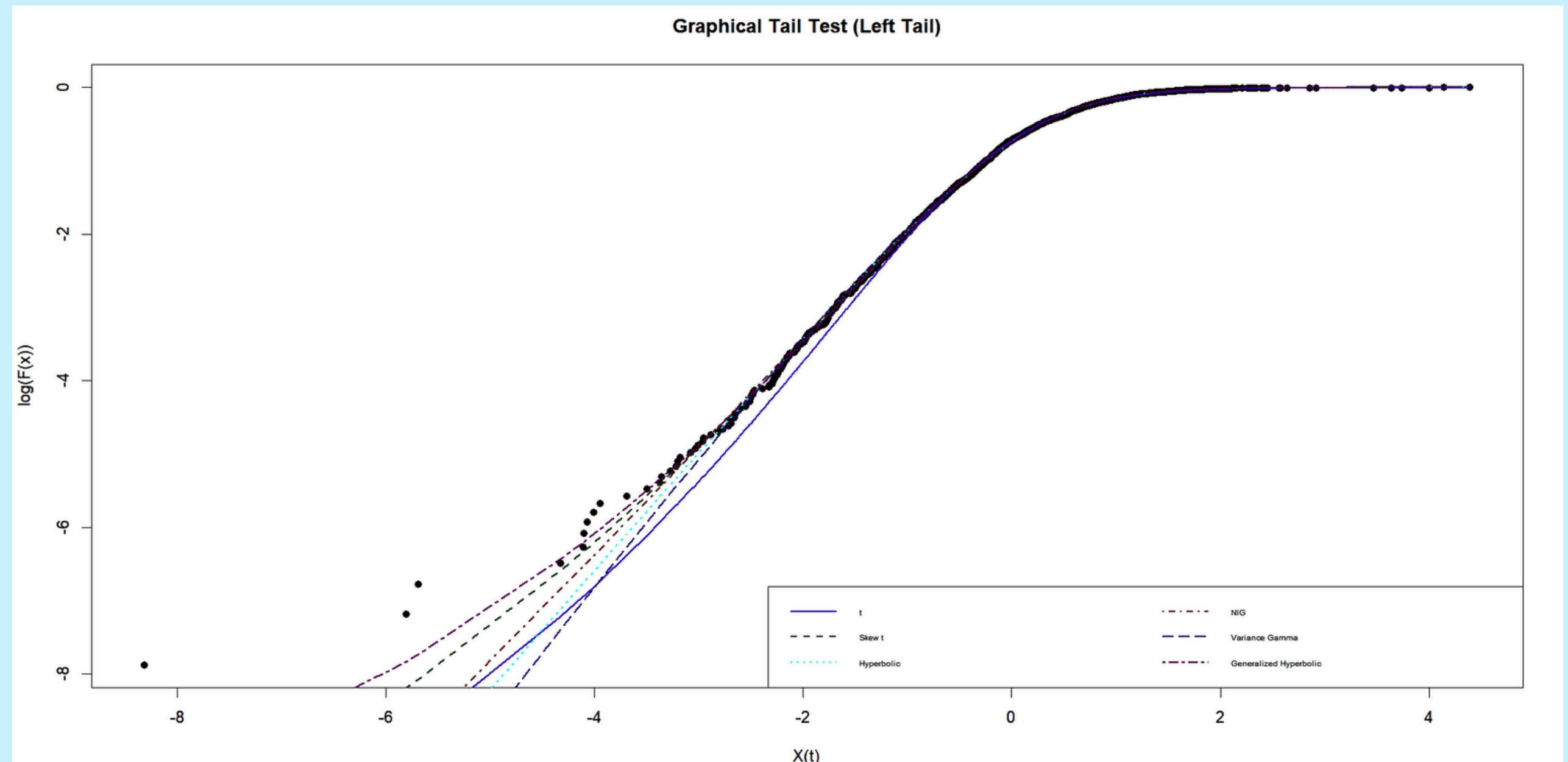
Model	Nifty50	Banknifty	Sensex
Normal Statistic p value	0.0481 ≤0.0000	0.04524 ≤0.0000	0.04813 ≤0.0000
Laplace Statistic p value	0.0475 ≤0.0000	0.03446 ≤0.0000	0.04040 ≤0.0000
Cauchy Statistic p value	0.0736 ≤0.0000	0.06364 ≤0.0000	0.07331 ≤0.0000
Student's t Statistic p value	0.0179 0.36552	0.00826 0.99712	0.04790 ≤0.0000
Skew normal Statistic p value	0.03198 0.00937	0.040999941 ≤0.0000	0.02858 ≤0.0000
Skew t Statistic p value	0.01670 0.45712	0.00846 0.9960	0.01698 0.43161

Model	Nifty50	Banknifty	Sensex
Skew Cauchy Statistic p value	0.0661 ≤0.0000	0.06118 ≤0.0000	0.06506 ≤0.0000
Skew Laplace Statistic p value	0.0481 ≤0.0000	0.03258 ≤0.0000	0.02314 0.11807
Hyperbolic Statistic p value	0.0177 0.3810	0.0107 0.9498	0.01763 0.38456
Normal inverse Gaussian Statistic p value	0.0173 0.4104	0.0110 0.9372	0.01726 0.41052
Variance gamma Statistic p value	0.0186 0.3214	0.0120 0.8874	0.01869 0.31465
Generalized hyperbolic Statistic p value	0.0156 0.5489	0.0080 0.9979	0.01721 0.41447

Methodology

6. Tail Modelling and Goodness of fit (Graphical Tail Test)

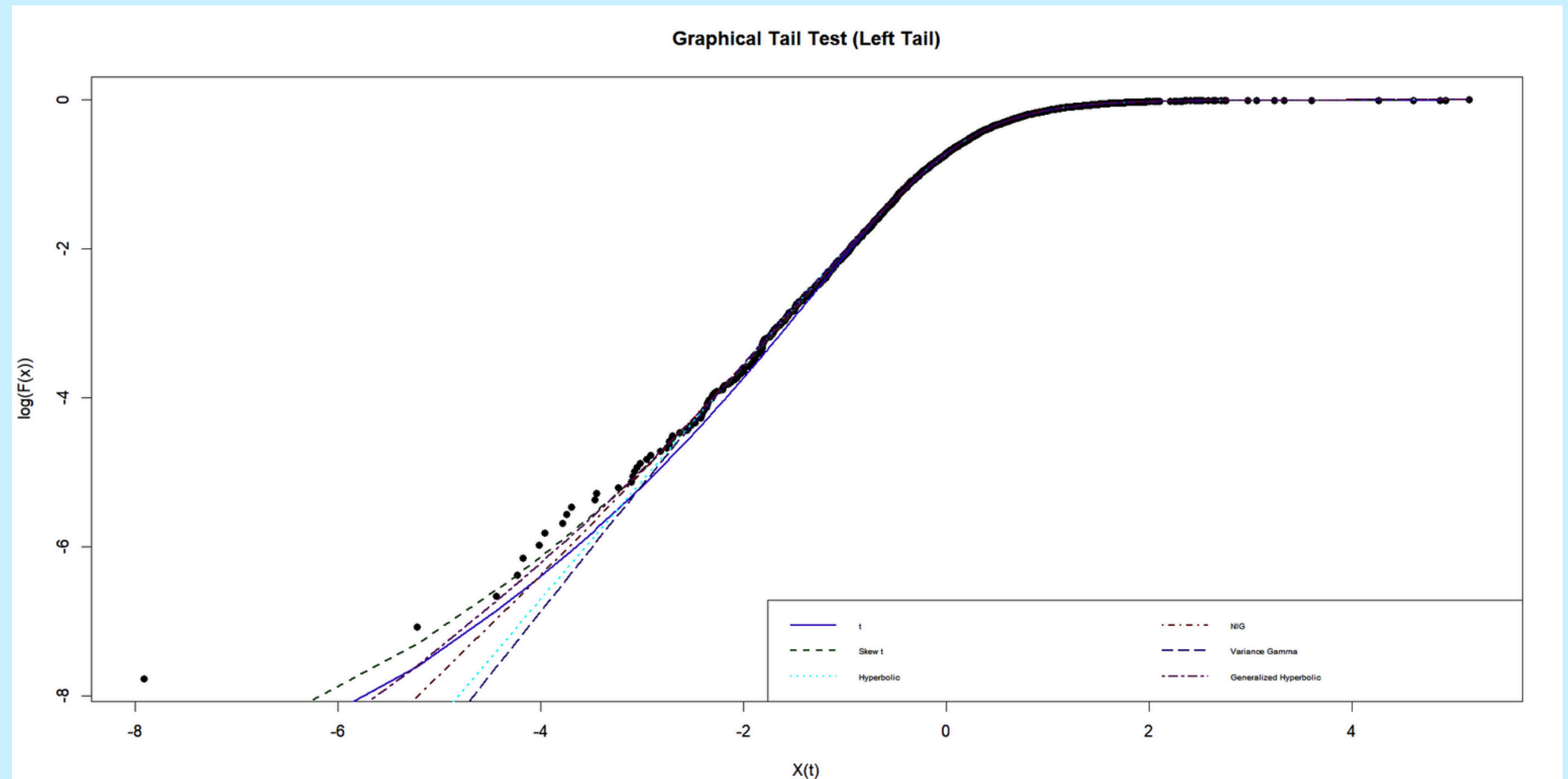
For Nifty 50:



Methodology

6. Tail Modelling and Goodness of fit (Graphical Tail Test)

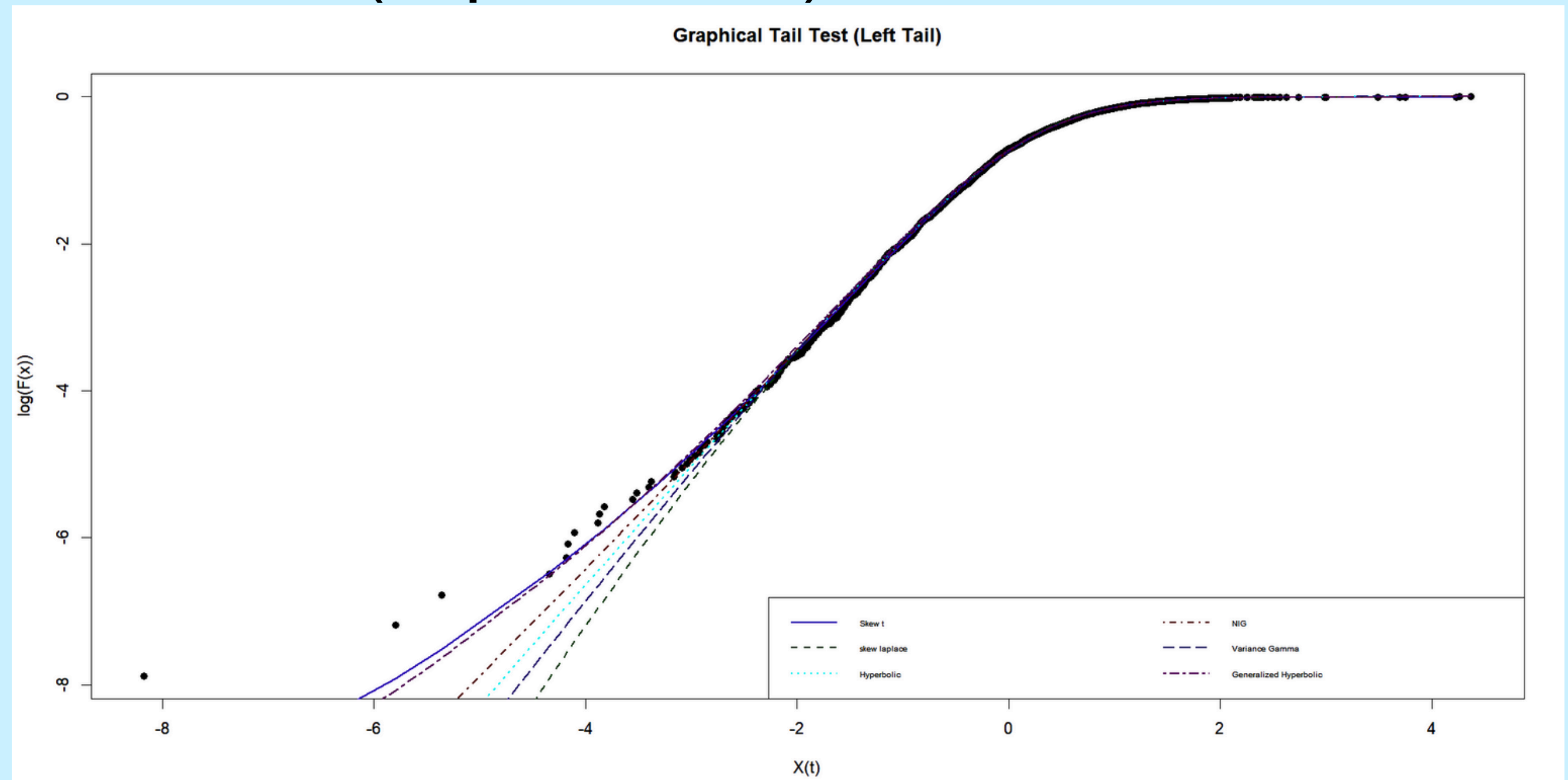
For Banknifty:



Methodology

6. Tail Modelling and Goodness of fit (Graphical Tail Test)

For Sensex:

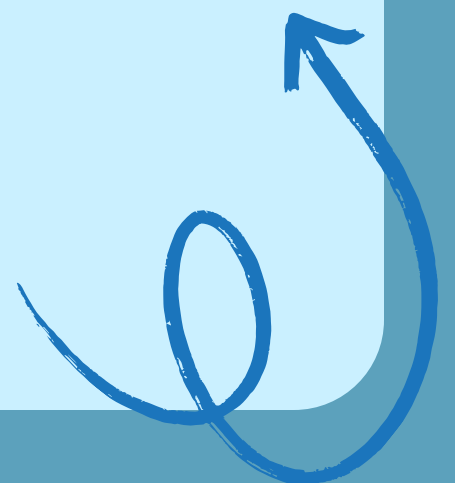


Methodology

6. Tail Modelling and Goodness of fit

By combining the results, the best probabilistic model for each stock index are as follows:

<i>STOCK INDEX</i>	<i>BEST FIT DISTRIBUTION</i>
<i>Nifty50</i>	<i>Generalized Hyperbolic</i>
<i>Banknifty</i>	<i>Skew-t</i>
<i>Sensex</i>	<i>Skew-t</i>



Methodology

7. Risk Measures (VaR and CTE)

1. VALUE-AT-RISK

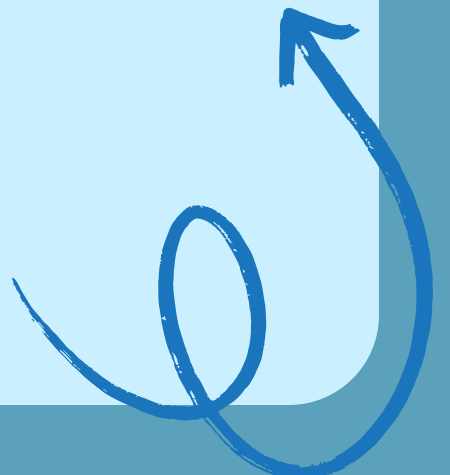
Value at Risk (VaR) estimates the maximum expected loss over a specific period at a chosen confidence level. It helps investors understand how much they could potentially lose under normal market conditions

$$VaR = \inf\{l \in \mathbb{R} : F_L(l) \geq \theta\}$$

2. CONDITIONAL TAIL EXPECTATION

Conditional Tail Expectation (CTE), also called Expected Shortfall, provides the average of losses that exceed the VaR threshold. It captures the severity of extreme losses that VaR alone may overlook. Thus, while VaR gives a cutoff level, CTE describes how large the losses can be beyond that cutoff.

$$CTE_{\theta} = E\{L | L > VaR_{\theta}\}$$



Methodology

8. Application to MLD payoff

The payoffs for Principal-Protected and Non-Principal Protected MLD's are defined as:

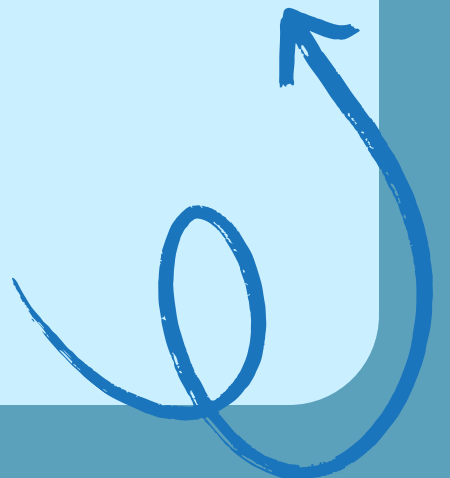
1. Principal Protected MLD:

$$Payoff = Principal(1 + \max(0, Prate \times returns))$$

2. Non-Principal Protected MLD:

$$Payoff = Principal (1 + Prate \times returns)$$

$$Loss = \frac{Principal - Payoff}{Principal}$$



Methodology

8. Application to MLD payoff (Using Historical Data)

- Direct Equity Investments
- Principal-Protected Market-linked debenture :
- Non- Principal-Protected Market-linked debenture:

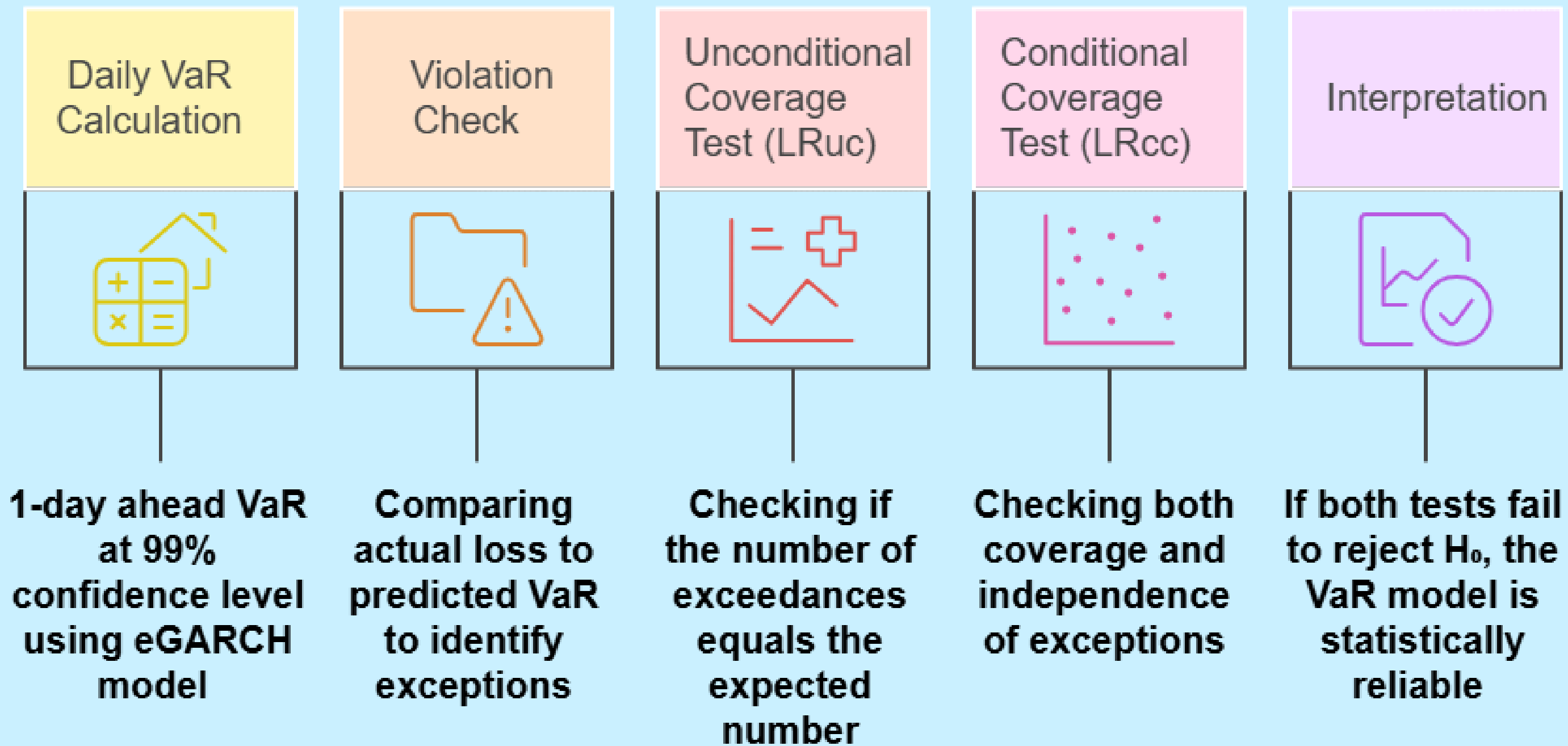
Measures	Nifty50 (%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.701793	4.182023	2.699757
VaR(95%)	1.497767	2.113891	1.474738
CTE(99%)	4.572059	6.689879	4.606388
CTE(95%)	2.454243	3.591946	2.452327

Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	0	0	0
VaR(95%)	0	0	0
CTE(99%)	0	0	0
CTE(95%)	0	0	0

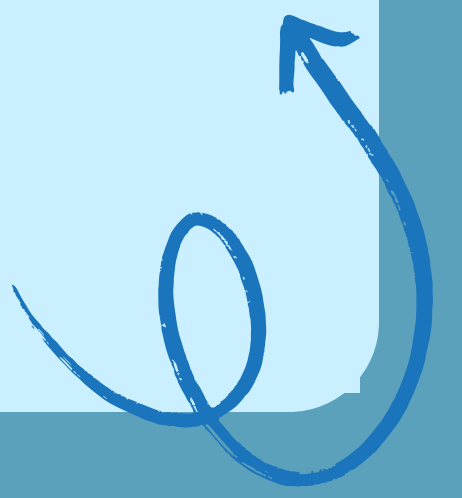
Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.161434	3.345618	2.159806
VaR(95%)	1.198213	1.691113	1.179791
CTE(99%)	3.657648	5.351903	3.68511
CTE(95%)	1.963395	2.873557	1.961861

Methodology

Backtesting of VaR Model



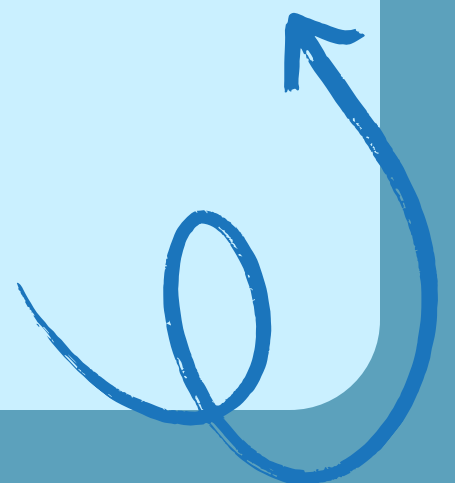
The 1-day ahead VaR is given as: $VaR_{t+1}(\hat{\theta}) = \hat{\mu}_{t+1} + q_{\theta} * \hat{\sigma}_{t+1}$



Methodology

Results of VaR Backtesting

Tests	Nifty50	Banknifty	Sensex
Expected Violations	26	23	26
Actual Violations	24	22	24
Unconditional Coverage test Statistic p value	0.1971162 0.6570587	0.09498533 0.7579325	0.2273146 0.6335222
Conditional Coverage test Statistic p value	0.6405406 0.7259528	0.5115148 0.7743298	0.6678561 0.7161053
Decision criteria	Do not Reject Ho ,for both test	Do not Reject Ho ,for both test	Do not Reject Ho ,for both test



Methodology

8. Application to MLD payoff (Using Monte-Carlo Simulation)

- Equity Direct Investment via simulation:
- Principal-Protected Market-linked debenture via simulation:
- Non- Principal-Protected Market-linked debenture via simulation:

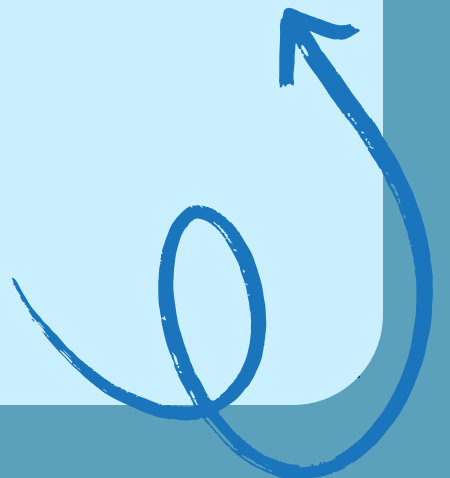
Measures	Nifty50 (%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.827698	2.682138	1.293541
VaR(95%)	1.663599	1.597946	0.763207
CTE(99%)	3.726811	3.521513	1.67347
CTE(95%)	2.413866	2.293143	1.1037623

Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	0	0	0
VaR(95%)	0	0	0
CTE(99%)	0	0	0
CTE(95%)	0	0	0

Measures	Nifty50(%)	Banknifty(%)	Sensex(%)
VaR(99%)	2.262158	2.145711	1.034833
VaR(95%)	1.330879	1.278357	0.6105656
CTE(99%)	2.981449	2.817211	1.338776
CTE(95%)	1.931093	1.834515	0.8830098

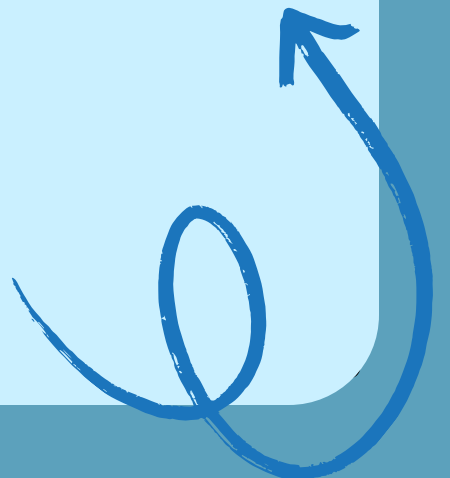
Conclusions:

1. For **Direct investments**, using both Empirical(historical data) and Monte Carlo Simulation ,it was found that Banknifty exhibits the highest risk followed by Nifty50 and Sensex.
2. For **Market-Linked Debentures**,
 - For Principal-Protected MLDs , both VaR and CTE were zero reflecting the feature complete capital protection in all scenarios.
 - Non-Principal-Protected MLDs ,showed moderate risk-levels, higher than Principal-Protected MLDs and lower than Direct-Equity investments .
 - Direct-Equity investments were found to be riskier because of complete exposure to market fluctuations especially in adverse market scenarios.
3. The parametric approach empowers investors and issuers in **Indian market-linked debentures**, offering enhanced risk quantification and informed decision-making, ultimately fostering **greater market transparency** and investor confidence.



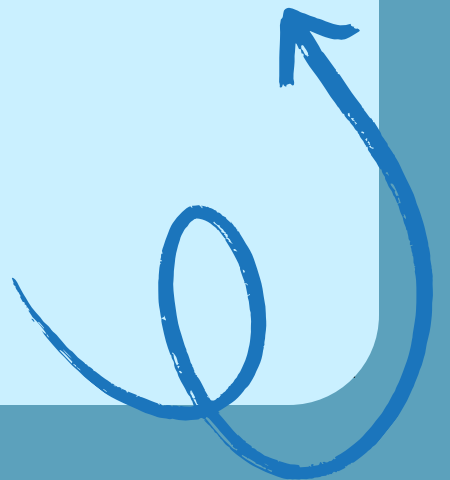
Limitations..

- Data Availability and Quality
- During market crisis, parametric models fail to capture extreme values
- The Indian MLD market is still developing, the heterogeneity in product design can make standardization challenging
- Parameters estimated may not remain stable over time due to policy changes
- Limited Consideration of Behavioral and Liquidity Factors



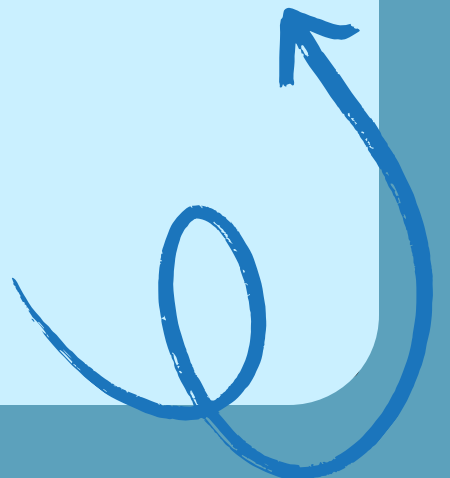
Future Scopes..

- Expansion to Machine Learning Approaches
- Integration of Credit and Liquidity Risk
- Dynamic and Regime-Switching Models
- Cross-Market and Cross-Product Analysis
- Development of a Standardized Risk Disclosure Framework
- Automation and Real-Time Application



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Thank You

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